AMT Worksheet 8
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The purpose of this worksheet is to give you some practice working with asymptotic expansions and expressions found using the method of steepest descents.

In problems 1 – 4, find the asymptotic expansion for the given integral expression for $f(x)$. Then, determine the number of terms that should be used in order to find the most precise value of $f(10)$ and use a computer algebra system to determine this most precise value along with its bounds (that is, find two neighboring approximations at the 'sweet spot' and give their average as the 'most precise value') and the maximum error associated with either of the two approximations. Finally, use a computer algebra system to perform the numerical integration and quote the error as the difference between this value and your 'most precise value'.

1) $f(x) = e^{2x}\int_x^\infty \frac{e^{-2t}}{t\sqrt{t}}dt$.

2) $f(x) = e^{x^2}\int_x^\infty \frac{e^{-t^2}}{t}dt$.

3) $f(x) = \int_0^\infty \frac{e^{-xt}}{1+t^2}dt$.

4) $f(x) = \int_0^\infty e^{-xt}\sqrt{1+t}\,dt$.

In problems 5 – 8, find an asymptotic expression for the given function which will be value when n is very large. Then, use a computer algebra system to compare the value of the integral and your expression for n = 5, 12, and 20. For larger values of n, which of these estimates do you trust more? Note: for problem 8, you may not be able to solve for the value of x you're trying to expand around exactly. Use a computer algebra system to find its approximate value instead. Use the symbol $x*$ in your expression.

5) $\int_0^\infty x^{n^2}n^{-x}dx$.

6) $\int_0^\infty e^{nx}x^{-x}dx$.

7) $\int_0^\infty n^{n\sqrt{x}}e^{-x}dx$.

8) $\int_0^\infty x^{n^2x}e^{-x^2}dx$.